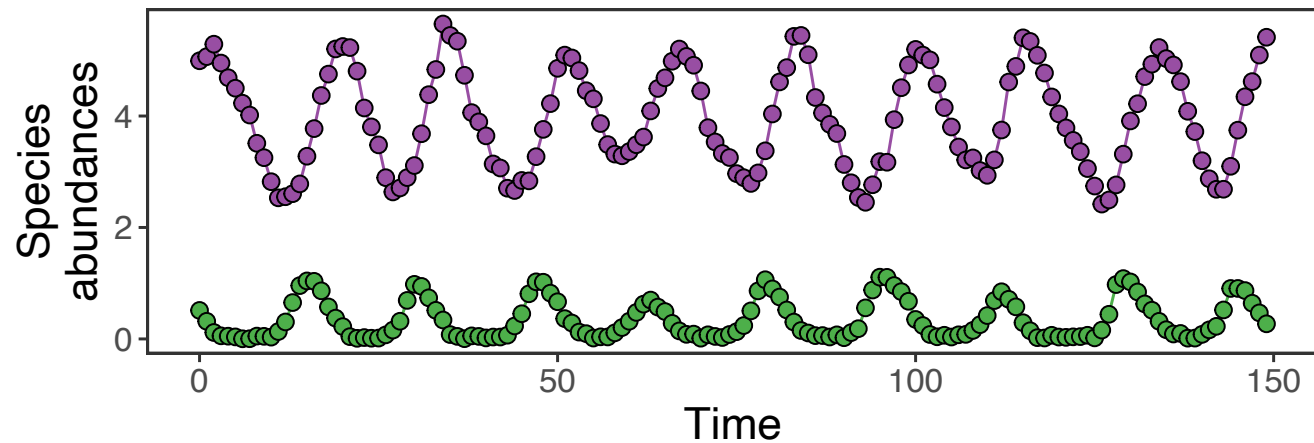
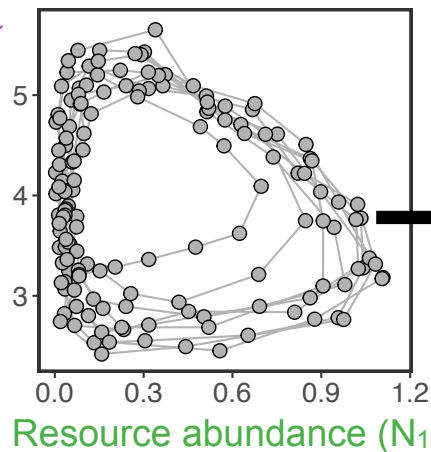
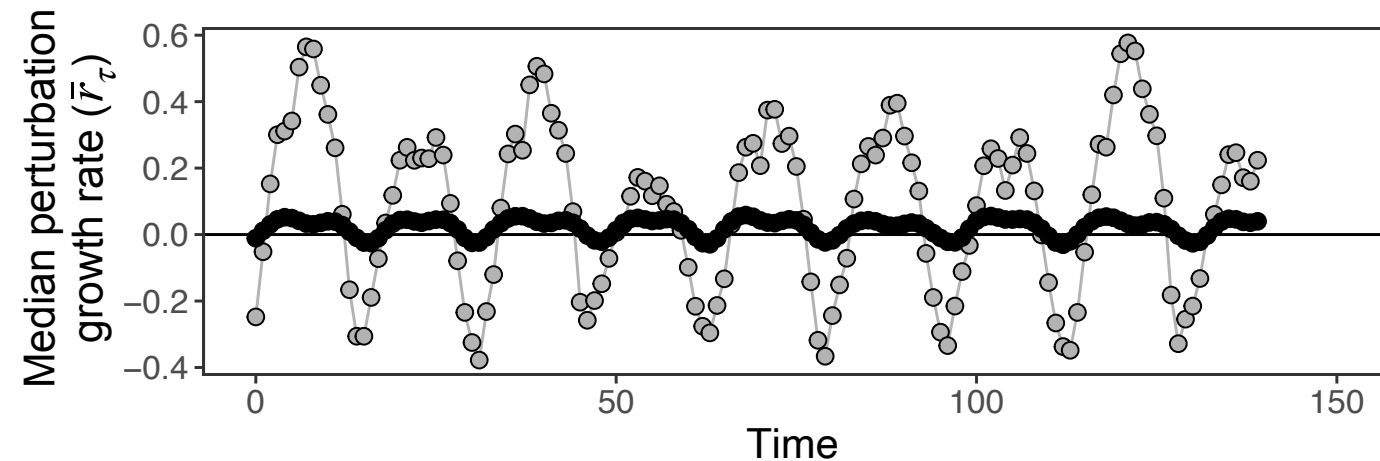


(a)Consumer abundance (N_2)

Time-series data
(2-species limit cycle with
process and observational noise)

(b)

○ True
● Inferred

Infer time-varying Jacobian matrix
by fitting a weighted regression
(S-map model) for each species

Compute time-varying stability
metric ($\min[r_\tau]$, $\bar{r}_\tau$, or $\max[r_\tau]$)
using matrix Φ for time scale τ